

Vehicle Dynamics, Battery, and Sensor Models (Dugoff Tire + Sensor Simulation)

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Abstract

This document presents the transition from a kinematic bicycle model to a dynamic vehicle model incorporating the Dugoff tire model with wheel rotational dynamics. We derive the complete force-based dynamics, normal load distribution with load transfer, slip ratio computation, friction-limited tire forces, wheel angular velocity integration, and Ackermann steering geometry. The hybrid torque-slip model enables realistic simulation of traction limits, wheel spin, wheel lock, and combined longitudinal-lateral dynamics for autonomous mining haul truck applications.

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1 Introduction

1.1 Motivation for Dynamic Model

The kinematic bicycle model assumes:

- **No slip:** Pure rolling condition at all times
- **Unlimited traction:** Tire forces can be arbitrarily large
- **Instantaneous response:** No force buildup dynamics

These assumptions break down when:

- Vehicle operates on low-friction surfaces (gravel, wet haul roads)
- Emergency braking or aggressive acceleration
- Combined maneuvers (braking while turning)
- Realistic traction control and ABS simulation required

The Dugoff tire model addresses these by:

- Computing friction-limited forces based on normal load
- Modeling slip ratios (longitudinal and lateral)
- Capturing smooth saturation from linear to sliding regimes

1.2 Model Architecture Evolution

We present three levels of modeling fidelity:

Kinematic Model (V1):

$$\text{Actuator} \rightarrow \text{Drive} \rightarrow \text{Velocity} \rightarrow \text{Position/Yaw} \quad (1)$$

Torque-Only Dynamic Model (V2):

$$\text{Actuator} \rightarrow \text{Drive} \rightarrow \boxed{\text{Tire Forces (friction-limited)}} \rightarrow \text{Acceleration} \rightarrow \text{Velocity} \quad (2)$$

Hybrid Torque-Slip Model (V2.5):

$$\text{Actuator} \rightarrow \boxed{\text{Wheel Dynamics}} \rightarrow \boxed{\text{Slip}} \rightarrow \boxed{\text{Dugoff Forces}} \rightarrow \text{Vehicle Dynamics} \quad (3)$$

The hybrid model adds wheel rotational dynamics, enabling natural slip development, wheel spin, and wheel lock simulation.

1.3 Model Comparison Summary

Feature	Kinematic	Torque-Only	Hybrid
Traction limit	Unlimited	μF_z	μF_z
Wheel slip	None	Estimated	Computed
Wheel spin	No	No	Yes
Wheel lock	No	No	Yes
TCS/ABS testing	No	Limited	Full
Computational cost	Low	Low	Medium

Table 1: Tire model comparison

2 Steering Geometry and Ackermann Model

2.1 Virtual Steering Angle

The bicycle model uses a single **virtual steering angle** δ to represent the combined effect of both front wheels. This simplification works well for trajectory planning but requires mapping to individual wheel angles for realistic tire force computation.

$$\delta = \text{virtual steering angle (bicycle model input)} \quad (4)$$

The virtual angle corresponds to the steering angle of a hypothetical single front wheel located at the center of the front axle.

2.2 Ackermann Steering Geometry

For a vehicle turning with radius R , all wheels must have their axes intersect at a common point (the Instantaneous Center of Rotation, ICR) to achieve pure rolling without tire scrub.

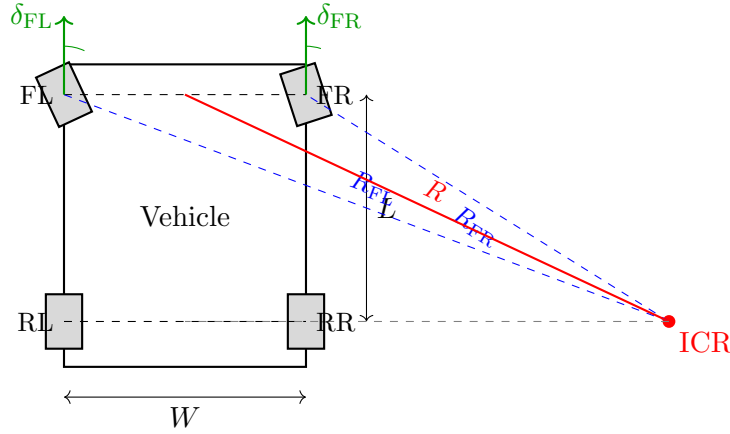


Figure 1: Ackermann steering geometry showing inner wheel (FR) turning more than outer wheel (FL) for a right turn

2.3 Ackermann Equations

Turn radius from virtual steering angle:

$$R = \frac{L}{\tan(\delta)} \quad (5)$$

where:

- R – turn radius from rear axle center to ICR (m)
- L – wheelbase (m)
- δ – virtual steering angle (rad)

Per-wheel turn radii:

For a right turn ($\delta > 0$), the turn center is to the right of the vehicle:

$$R_{FL} = R + \frac{W}{2} \quad (\text{outer wheel, larger radius}) \quad (6)$$

$$R_{FR} = R - \frac{W}{2} \quad (\text{inner wheel, smaller radius}) \quad (7)$$

where W is the track width (m).

Individual wheel steering angles:

$$\delta_{\text{FL}} = \arctan\left(\frac{L}{R + W/2}\right) \quad (8)$$

$$\delta_{\text{FR}} = \arctan\left(\frac{L}{R - W/2}\right) \quad (9)$$

Key property: For a right turn, the inner wheel (FR) turns *more* than the outer wheel (FL):

$$|\delta_{\text{FR}}| > |\delta_{\text{FL}}| \quad \text{when } \delta > 0 \quad (10)$$

This ensures all wheel axes intersect at the ICR, minimizing tire scrub.

2.4 Ackermann Condition

The classic Ackermann condition states:

$$\cot(\delta_{\text{outer}}) - \cot(\delta_{\text{inner}}) = \frac{W}{L} \quad (11)$$

Proof: From the geometry:

$$\cot(\delta_{\text{FL}}) = \frac{R + W/2}{L} \quad (12)$$

$$\cot(\delta_{\text{FR}}) = \frac{R - W/2}{L} \quad (13)$$

$$\cot(\delta_{\text{FL}}) - \cot(\delta_{\text{FR}}) = \frac{(R + W/2) - (R - W/2)}{L} = \frac{W}{L} \quad \square \quad (14)$$

This condition is independent of turn radius R and depends only on vehicle geometry.

2.5 Ackermann Percentage

Real vehicles rarely use perfect (100%) Ackermann geometry. A partial Ackermann configuration blends between:

- **Full Ackermann (100%):** $\delta_{\text{FL}} \neq \delta_{\text{FR}}$ as derived above
- **Parallel steering (0%):** $\delta_{\text{FL}} = \delta_{\text{FR}} = \delta$

The Ackermann percentage $\alpha_{\text{ack}} \in [0, 1]$ controls the blend:

$$\delta_{\text{FL}} = \alpha_{\text{ack}} \cdot \delta_{\text{FL,Ackermann}} + (1 - \alpha_{\text{ack}}) \cdot \delta \quad (15)$$

$$\delta_{\text{FR}} = \alpha_{\text{ack}} \cdot \delta_{\text{FR,Ackermann}} + (1 - \alpha_{\text{ack}}) \cdot \delta \quad (16)$$

Typical values:

- Passenger cars: $\alpha_{\text{ack}} = 0.80 - 0.90$ (compromise for tire wear at speed)
- Racing vehicles: $\alpha_{\text{ack}} = 0.50 - 0.70$ (optimized for cornering)
- Mining trucks: $\alpha_{\text{ack}} = 0.95 - 1.00$ (low-speed maneuverability)

2.6 Speed-Dependent Steering Limit

At high speeds, large steering angles cause excessive lateral acceleration and potential rollover. A speed-dependent limit reduces maximum steering:

$$\delta_{\max}(v) = \delta_{\max,0} \cdot r(v) \quad (17)$$

where the reduction factor $r(v)$ is:

$$r(v) = \begin{cases} 1 & v \leq v_{\text{start}} \\ 1 - (1 - r_{\text{high}}) \cdot \frac{v - v_{\text{start}}}{v_{\text{end}} - v_{\text{start}}} & v_{\text{start}} < v < v_{\text{end}} \\ r_{\text{high}} & v \geq v_{\text{end}} \end{cases} \quad (18)$$

Parameters:

- $\delta_{\max,0}$ – maximum steering angle at low speed (rad)
- v_{start} – speed at which limiting begins (m/s)
- v_{end} – speed at which full limiting is applied (m/s)
- r_{high} – reduction ratio at high speed (e.g., $0.35 = 35\%$ of max)

Example for XCMG XDE320:

$$\delta_{\max,0} = 35 \quad (19)$$

$$v_{\text{start}} = 8 \text{ m/s} \quad (20)$$

$$v_{\text{end}} = 25 \text{ m/s} \quad (21)$$

$$r_{\text{high}} = 0.35 \quad (22)$$

At 25 m/s (90 km/h), maximum steering is limited to $35 \times 0.35 = 12.25$.

2.7 Steering Rate Limiting

Physical steering systems have bandwidth limitations due to hydraulic flow (mining trucks) or electric motor limits (passenger cars):

$$\dot{\delta} = \text{clamp} \left(\frac{\delta_{\text{cmd}} - \delta}{\Delta t}, -\dot{\delta}_{\max}, +\dot{\delta}_{\max} \right) \quad (23)$$

Typical maximum steering rates:

- Passenger cars (electric): $\dot{\delta}_{\max} = 400 - 600 \text{ }^\circ/\text{s}$
- Mining trucks (hydraulic): $\dot{\delta}_{\max} = 30 - 60 \text{ }^\circ/\text{s}$

The discrete-time update becomes:

$$\delta[k+1] = \delta[k] + \text{clamp} \left(\delta_{\text{des}} - \delta[k], -\dot{\delta}_{\max}\Delta t, +\dot{\delta}_{\max}\Delta t \right) \quad (24)$$

2.8 Minimum Turn Radius

At maximum steering angle, the minimum turn radius is:

$$R_{\min} = \frac{L}{\tan(\delta_{\max})} \quad (25)$$

For XCMG XDE320 with $L = 6.3$ m and $\delta_{\max} = 35$:

$$R_{\min} = \frac{6.3}{\tan(35)} = \frac{6.3}{0.700} = 9.0 \text{ m} \quad (26)$$

The inner wheel turn radius is:

$$R_{\text{inner}} = R_{\min} - \frac{W}{2} = 9.0 - 2.0 = 7.0 \text{ m} \quad (27)$$

This determines the minimum pit road width for U-turns.

2.9 Steering System Block Diagram

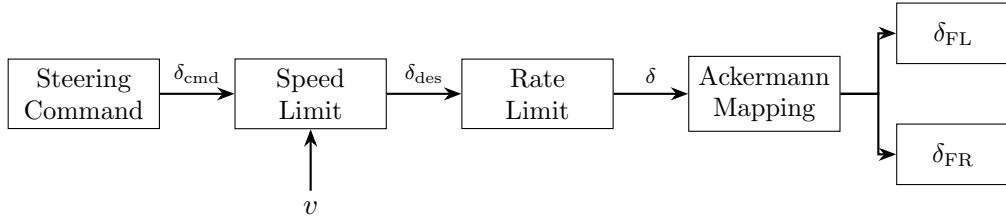


Figure 2: Steering system signal flow

3 Force-Based Longitudinal Dynamics

3.1 Equations of Motion

Newton's second law in the longitudinal direction:

$$m \frac{dv}{dt} = \sum F_x - F_{\text{drag}} - F_{\text{roll}} \quad (28)$$

where:

$$\sum F_x = F_{x,\text{fl}} + F_{x,\text{fr}} + F_{x,\text{rl}} + F_{x,\text{rr}} \quad (\text{sum of tire forces}) \quad (29)$$

$$F_{\text{drag}} = \frac{1}{2} \rho C_d A v^2 \quad (30)$$

$$F_{\text{roll}} = C_{\text{roll}} \cdot \text{sgn}(v) \quad (31)$$

3.2 Tire Force Allocation

For a rear-wheel-drive vehicle:

$$F_{x,\text{fl}} = 0 \quad (\text{front wheels: unpowered}) \quad (32)$$

$$F_{x,\text{fr}} = 0 \quad (33)$$

$$F_{x,\text{rl}} = f_{\text{Dugoff}}(\sigma_{x,\text{rl}}, \sigma_{y,\text{rl}}, F_{z,\text{rl}}) \quad (34)$$

$$F_{x,\text{rr}} = f_{\text{Dugoff}}(\sigma_{x,\text{rr}}, \sigma_{y,\text{rr}}, F_{z,\text{rr}}) \quad (35)$$

where f_{Dugoff} is the tire model function (detailed in Section 5).

4 Normal Load Distribution

4.1 Static Load Distribution

Weight distribution between front and rear axles:

$$F_{z,\text{front}}^{\text{static}} = \frac{mg \cdot L_r}{L} \quad (36)$$

$$F_{z,\text{rear}}^{\text{static}} = \frac{mg \cdot L_f}{L} \quad (37)$$

where:

- m – vehicle mass (kg)
- $g = 9.81 \text{ m/s}^2$ – gravitational acceleration
- $L = L_f + L_r$ – wheelbase (m)
- L_f – distance from CoG to front axle (m)
- L_r – distance from CoG to rear axle (m)

4.2 Longitudinal Load Transfer

During acceleration or braking, normal load shifts between axles:

$$\boxed{\Delta F_z = \frac{m \cdot a_x \cdot h_{\text{cg}}}{L}} \quad (38)$$

where:

- a_x – longitudinal acceleration (m/s^2), positive = acceleration
- h_{cg} – center of gravity height (m)

Dynamic normal loads:

$$F_{z,\text{front}} = F_{z,\text{front}}^{\text{static}} - \Delta F_z \quad (39)$$

$$F_{z,\text{rear}} = F_{z,\text{rear}}^{\text{static}} + \Delta F_z \quad (40)$$

Physical interpretation:

- $a_x > 0$ (acceleration): Weight transfers to rear $\Rightarrow$ more rear traction
- $a_x < 0$ (braking): Weight transfers to front $\Rightarrow$ more front braking capacity

4.3 Lateral Load Transfer

During cornering with lateral acceleration a_y :

$$\Delta F_{z,\text{lat}} = \frac{m \cdot a_y \cdot h_{\text{cg}}}{W} \quad (41)$$

where W is the track width (m).

4.4 Per-Wheel Normal Loads

Combined longitudinal and lateral load transfer:

$$F_{z,fl} = \frac{1}{2} (F_{z,front} - \Delta F_{z,lat,front}) \quad (42)$$

$$F_{z,fr} = \frac{1}{2} (F_{z,front} + \Delta F_{z,lat,front}) \quad (43)$$

$$F_{z,rl} = \frac{1}{2} (F_{z,rear} - \Delta F_{z,lat,rear}) \quad (44)$$

$$F_{z,rr} = \frac{1}{2} (F_{z,rear} + \Delta F_{z,lat,rear}) \quad (45)$$

4.5 Example: XCMG XDE320 Mining Truck

Parameters:

$$m = 218,000 \text{ kg (empty)} \quad (46)$$

$$L = 6.3 \text{ m} \quad (47)$$

$$h_{cg} = 2.8 \text{ m} \quad (48)$$

$$L_f = 2.52 \text{ m (40\% of wheelbase)} \quad (49)$$

$$L_r = 3.78 \text{ m (60\% of wheelbase)} \quad (50)$$

Static loads:

$$F_{z,front}^{static} = \frac{218,000 \times 9.81 \times 3.78}{6.3} = 1,284,000 \text{ N} \quad (51)$$

$$F_{z,rear}^{static} = \frac{218,000 \times 9.81 \times 2.52}{6.3} = 856,000 \text{ N} \quad (52)$$

Load transfer during braking at $a_x = -2.0 \text{ m/s}^2$:

$$\Delta F_z = \frac{218,000 \times (-2.0) \times 2.8}{6.3} = -193,600 \text{ N} \quad (53)$$

$$F_{z,front} = 1,284,000 + 193,600 = 1,478,000 \text{ N} \quad (54)$$

$$F_{z,rear} = 856,000 - 193,600 = 662,000 \text{ N} \quad (55)$$

Result: Front axle load increases by 15%, rear decreases by 23% during braking.

5 Dugoff Tire Model

5.1 Fundamental Equations

Tire forces are given by:

$$\boxed{F_x = C_x \sigma_x f(\lambda)} \quad (56)$$

$$\boxed{F_y = C_y \sigma_y f(\lambda)} \quad (57)$$

where:

- F_x, F_y – longitudinal and lateral tire forces (N)
- C_x, C_y – slip stiffness coefficients (N), load-dependent
- σ_x, σ_y – longitudinal and lateral slip ratios
- $f(\lambda)$ – friction saturation function
- λ – friction utilization parameter

5.2 Slip Ratios

Longitudinal slip ratio:

$$\sigma_x = \frac{\omega R - V_x}{\max(|V_x|, |\omega R|, V_{\min})} \quad (58)$$

where:

- ω – wheel angular velocity (rad/s)
- R – tire effective radius (m)
- V_x – vehicle longitudinal velocity (m/s)
- $V_{\min} \approx 0.1$ m/s – prevents division by zero

Sign convention:

$$\sigma_x > 0 \quad \text{Driving (wheel faster than vehicle)} \quad (59)$$

$$\sigma_x < 0 \quad \text{Braking (wheel slower than vehicle)} \quad (60)$$

$$\sigma_x = 0 \quad \text{Free rolling} \quad (61)$$

Lateral slip ratio:

$$\sigma_y = \frac{V_y}{V_x} \approx \tan(\alpha) \approx \alpha \quad (62)$$

where α is tire slip angle (rad), valid for small angles ($|\alpha| < 0.2$ rad).

5.3 Load-Dependent Slip Stiffness

Tire stiffness scales with normal load:

$$C_x(F_z) = C_{x,\text{base}} \cdot \frac{F_z}{F_{z,\text{ref}}} \quad (63)$$

$$C_y(F_z) = C_{y,\text{base}} \cdot \frac{F_z}{F_{z,\text{ref}}} \quad (64)$$

where:

- $C_{x,\text{base}}, C_{y,\text{base}}$ – baseline stiffness at reference load (N)
- $F_{z,\text{ref}}$ – reference normal load (N)

Typical values for mining truck tires:

- $C_{x,\text{base}} \approx 280,000$ N
- $C_{y,\text{base}} \approx 220,000$ N
- $F_{z,\text{ref}} = 100,000$ N

5.4 Friction Utilization Parameter

The key parameter governing saturation:

$$\lambda = \frac{\mu F_z}{2\sqrt{(C_x \sigma_x)^2 + (C_y \sigma_y)^2}} \quad (65)$$

where μ is the coefficient of friction (surface-dependent).

Physical interpretation:

$$\lambda \geq 1 \quad \text{Linear regime (tire not saturated)} \quad (66)$$

$$\lambda < 1 \quad \text{Nonlinear regime (friction-limited, sliding)} \quad (67)$$

5.5 Friction Saturation Function

$$f(\lambda) = \begin{cases} (2 - \lambda)\lambda, & \lambda < 1 \\ 1, & \lambda \geq 1 \end{cases} \quad (68)$$

Properties:

- Continuous at $\lambda = 1$: $\lim_{\lambda \rightarrow 1^-} f(\lambda) = 1$
- Smooth transition from linear to saturated regime
- Maximum value: $f_{\max} = 1$ at $\lambda = 1$

5.6 Friction Circle Constraint

Theorem: The total tire force never exceeds the friction limit:

$$\sqrt{F_x^2 + F_y^2} \leq \mu F_z \quad (69)$$

Proof: For $\lambda < 1$ (friction-limited case):

$$\sqrt{F_x^2 + F_y^2} = \sqrt{(C_x \sigma_x f)^2 + (C_y \sigma_y f)^2} \quad (70)$$

$$= f(\lambda) \sqrt{C_x^2 \sigma_x^2 + C_y^2 \sigma_y^2} \quad (71)$$

$$= (2\lambda - \lambda^2) \cdot \frac{\mu F_z}{2\lambda} \quad (\text{from definition of } \lambda) \quad (72)$$

$$= \left(1 - \frac{\lambda}{2}\right) \mu F_z \quad (73)$$

$$< \mu F_z \quad (\text{since } 0 < \lambda < 1) \quad \square \quad (74)$$

5.7 Slip-Force Characteristic Curve

The Dugoff model produces the characteristic slip-force curve:

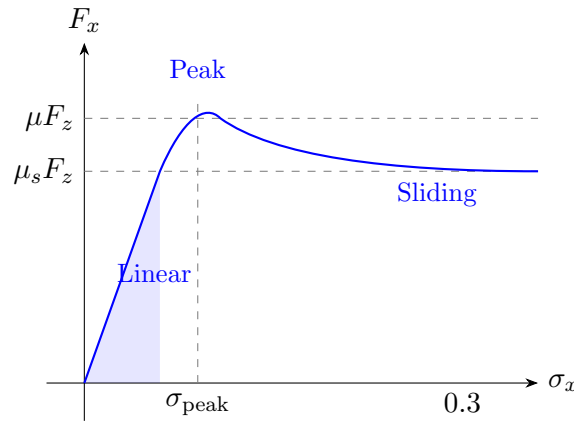


Figure 3: Dugoff slip-force characteristic showing linear region, peak grip, and sliding region

6 Wheel Rotational Dynamics

This section introduces the key addition for the hybrid model: independent wheel speed integration.

6.1 Motivation

In the torque-only model, wheel speed is derived from vehicle speed:

$$\omega = \frac{V_x}{R} \Rightarrow \sigma_x = 0 \text{ always!} \quad (75)$$

This prevents modeling of:

- Wheel spin during aggressive acceleration
- Wheel lock during hard braking
- Natural slip development
- TCS/ABS algorithm testing

The solution is to integrate wheel angular velocity independently.

6.2 Single Wheel Free Body Diagram

For each wheel, applying Newton's second law for rotation:

$$\boxed{I_w \dot{\omega} = \tau_{\text{drive}} - \tau_{\text{brake}} - F_x \cdot R} \quad (76)$$

where:

- I_w = wheel rotational inertia ($\text{kg}\cdot\text{m}^2$)
- ω = wheel angular velocity (rad/s)
- $\dot{\omega}$ = wheel angular acceleration (rad/s^2)
- τ_{drive} = drive torque at wheel ($\text{N}\cdot\text{m}$)
- τ_{brake} = brake torque ($\text{N}\cdot\text{m}$), always ≥ 0
- F_x = longitudinal tire force from Dugoff (N)
- R = tire effective rolling radius (m)

6.3 Drive Torque Distribution

For a rear-wheel drive (RWD) configuration:

$$\tau_{\text{drive,rl}} = \tau_{\text{drive,rr}} = \frac{\tau_{\text{motor}} \cdot N_{\text{gear}} \cdot \eta_{\text{drivetrain}}}{2} \quad (77)$$

$$\tau_{\text{drive,fl}} = \tau_{\text{drive,fr}} = 0 \quad (\text{front wheels not driven}) \quad (78)$$

where:

- τ_{motor} = motor shaft torque ($\text{N}\cdot\text{m}$)
- N_{gear} = final drive gear ratio (dimensionless)
- $\eta_{\text{drivetrain}}$ = drivetrain efficiency (typically 0.90-0.95)

6.4 Brake Torque Distribution

Brake torque is distributed with a front/rear bias:

$$\tau_{\text{brake,fl}} = \tau_{\text{brake,fr}} = \frac{\tau_{\text{brake,total}} \cdot \beta_{\text{front}}}{2} \quad (79)$$

$$\tau_{\text{brake,rl}} = \tau_{\text{brake,rr}} = \frac{\tau_{\text{brake,total}} \cdot (1 - \beta_{\text{front}})}{2} \quad (80)$$

where β_{front} is the front brake bias (typically 0.40-0.60).

For mining trucks with rear weight bias: $\beta_{\text{front}} = 0.40$ (40% front, 60% rear).

6.5 Wheel Inertia

The wheel rotational inertia includes:

$$I_w = I_{\text{tire}} + I_{\text{rim}} + I_{\text{brake}} + I_{\text{hub}} \quad (81)$$

Typical values:

- Passenger car: $I_w \approx 1.5 - 3.0 \text{ kg}\cdot\text{m}^2$
- Mining haul truck (XCMG XDE320): $I_w \approx 800 - 1500 \text{ kg}\cdot\text{m}^2$

6.6 Discrete-Time Integration

Using forward Euler integration:

$$\omega[k+1] = \omega[k] + \frac{\Delta t}{I_w} (\tau_{\text{drive}} - \tau_{\text{brake}} \cdot \text{sgn}(\omega[k]) - F_x[k] \cdot R) \quad (82)$$

Wheel Lock Detection:

$$\text{if } \omega[k] > 0 \text{ and } \omega[k+1] < 0 \text{ and } \tau_{\text{brake}} > 0 : \quad \omega[k+1] = 0 \quad (83)$$

This prevents non-physical sign reversal during braking.

6.7 Stability Criterion

The explicit integration is stable when:

$$\Delta t < \frac{2I_w}{C_x R^2} \quad (84)$$

For XCMG XDE320 parameters ($I_w = 1000 \text{ kg}\cdot\text{m}^2$, $C_x = 280,000 \text{ N}$, $R = 1.93 \text{ m}$):

$$\Delta t_{\text{max}} \approx \frac{2 \times 1000}{280,000 \times 1.93^2} \approx 1.9 \text{ ms} \quad (85)$$

Simulation timestep of 1 ms provides adequate stability margin.

7 Battery and Energy Model

This section documents the battery state and power-flow logic as implemented in **BatteryPlant** and used by the drivetrain/drive subsystem.

7.1 State Variables

The battery plant tracks:

- State of charge: $\text{SOC} \in [0, 1]$
- Terminal voltage: V (V)
- Current: I (A), with the convention $I > 0 = \text{discharge}$, $I < 0 = \text{charge (regen)}$
- Electrical power: P (W), with $P > 0 = \text{discharge}$, $P < 0 = \text{charge}$

7.2 Open-Circuit Voltage (SOC-Dependent)

A simple SOC-dependent voltage model is used (updated *before* computing current):

$$V(\text{SOC}) = V_{\text{nom}} (0.85 + 0.30 \text{SOC}) \quad (86)$$

with $V_{\text{nom}} = 400$ V. This yields 340 V at 0% SOC, 400 V at 50%, and 460 V at 100%.

7.3 Power Clamping to Motor Limits

The battery plant receives an electrical power demand (from the drive plant). This demand is clamped to the motor power capability:

$$P_{\text{dem}} \leftarrow \text{clamp}(P_{\text{dem}}, -P_{\text{max}}, +P_{\text{max}}) \quad (87)$$

where $P_{\text{max}} = P_{\text{motor,max}}$.

7.4 Charge/Discharge Efficiency Selection

Different efficiencies are applied depending on direction of energy flow:

$$\eta = \begin{cases} \eta_{\text{dis}} & \text{if } P_{\text{dem}} > 0 \quad (\text{discharge}) \\ \eta_{\text{chg}} & \text{if } P_{\text{dem}} \leq 0 \quad (\text{charge}) \end{cases} \quad (88)$$

7.5 Discrete SOC Update (Energy in Wh)

Energy over a timestep Δt is computed in Watt-hours:

$$\Delta E_{\text{Wh}} = \frac{P_{\text{dem}} \Delta t}{3600} \quad (89)$$

$$C_{\text{Wh}} = 1000 C_{\text{kWh}} \quad (90)$$

and SOC is updated as:

$$\Delta \text{SOC} = - \left(\frac{\Delta E_{\text{Wh}}}{C_{\text{Wh}}} \right) \eta, \quad \text{SOC}_{k+1} = \text{clamp}(\text{SOC}_k + \Delta \text{SOC}, \text{SOC}_{\text{min}}, \text{SOC}_{\text{max}}) \quad (91)$$

7.6 Electrical Power and Current

A regenerative power term P_{regen} (tracked from the regen logic) is applied as:

$$P = P_{\text{dem}} - P_{\text{regen}} \quad (92)$$

and the terminal current is:

$$I = \frac{P}{V(\text{SOC})}. \quad (93)$$

7.7 DrivePlant Coupling

In the drive subsystem, the motor power demand is computed from motor torque and motor speed:

$$P_{\text{motor}} = \tau_{\text{motor}} \omega_{\text{motor}}, \quad P_{\text{dem,kW}} = \frac{P_{\text{motor}}}{1000}. \quad (94)$$

Battery energy flow is then handled in three regimes (when the system is enabled):

- **Driving (consumption):** if $P_{\text{dem,kW}} > 0$, energy consumed is $\Delta E = P_{\text{dem,kW}} \Delta t \cdot 1000$ (J) and SOC decreases via `consume_energy()`.
- **Active regen (commanded negative torque):** if $P_{\text{dem,kW}} < 0$, the recovered electrical power is

$$P_{\text{regen,kW}} = -P_{\text{dem,kW}} \eta_{\text{regen,active}} \quad (95)$$

and SOC increases via `store_energy()`.

- **Coasting regen (resistive losses):** when braking is near zero and $|v|$ is above standstill threshold, a small recovery proportional to resistive forces is modeled:

$$P_{\text{regen,kW}} = \frac{(F_{\text{drag}} + F_{\text{roll}}) |v|}{1000} \eta_{\text{regen,coast}}, \quad F_{\text{drag}} = c_d v |v|. \quad (96)$$

Note: The detailed friction-limited wheel/tire dynamics determine whether negative motor torque translates into actual wheel deceleration and thus realistic regen behavior.

8 Closed-Loop System Architecture

8.1 System Block Diagram

The hybrid model forms a closed-loop system:

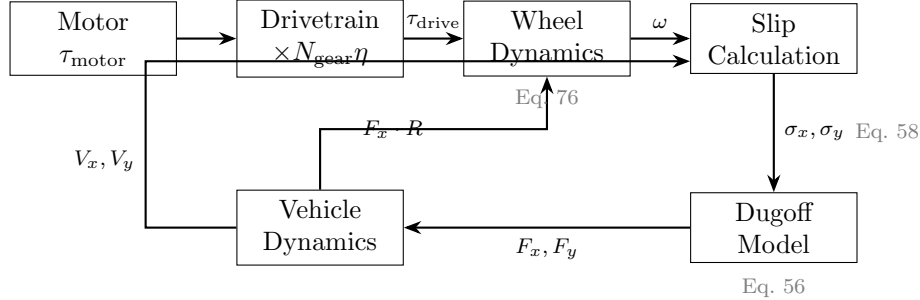


Figure 4: Closed-loop system architecture for hybrid torque-slip model

8.2 Causality Chain

At each timestep k :

1. **Input:** Motor torque command τ_{motor}
2. **Drivetrain:** $\tau_{\text{wheel}} = \tau_{\text{motor}} \cdot N_{\text{gear}} \cdot \eta$
3. **Wheel dynamics:** Integrate $\omega[k+1]$ using Eq. 82
4. **Slip calculation:** Compute σ_x, σ_y using Eq. 58
5. **Dugoff model:** Compute F_x, F_y using Eq. 56, 57

6. **Vehicle dynamics:** Update V_x, V_y from tire forces
7. **Feedback:** Tire reaction torque $F_x \cdot R$ affects next wheel integration

8.3 Key Insight: Feedback Loop

The tire force F_x appears on **both sides** of the system:

- **Output:** Accelerates the vehicle ($m\dot{V}_x = \sum F_x$)
- **Feedback:** Decelerates the wheel ($I_w\dot{\omega} = \tau - F_x R$)

This creates natural self-regulation:

- High slip $\rightarrow$ reduced F_x (saturation) $\rightarrow$ wheel accelerates less
- Low slip $\rightarrow$ high $F_x \rightarrow$ wheel slows toward free-rolling

9 Lateral Velocity and Slip Angle Computation

9.1 Front Axle Lateral Velocity

Lateral velocity at front tire contact patch:

$$V_{y,\text{front}} = V_y + \dot{\psi} \cdot L_f - V_x \tan(\delta) \quad (97)$$

where:

- V_y – vehicle lateral velocity at CoG (m/s)
- $\dot{\psi}$ – yaw rate (rad/s)
- δ – steering angle (rad)

9.2 Rear Axle Lateral Velocity

$$V_{y,\text{rear}} = V_y - \dot{\psi} \cdot L_r \quad (98)$$

9.3 Per-Wheel Lateral Slip Ratio

For front wheels (with Ackermann steering):

$$\sigma_{y,\text{fl}} = \frac{V_y + \dot{\psi} L_f - V_x \tan(\delta_{\text{FL}})}{V_x} \quad (99)$$

$$\sigma_{y,\text{fr}} = \frac{V_y + \dot{\psi} L_f - V_x \tan(\delta_{\text{FR}})}{V_x} \quad (100)$$

For rear wheels:

$$\sigma_{y,\text{rl}} = \sigma_{y,\text{rr}} = \frac{V_y - \dot{\psi} L_r}{V_x} \quad (101)$$

10 Torque-Only vs Hybrid Model

10.1 Torque-Only Model (Current Default)

Direct force computation without wheel dynamics:

$$F_x = \min \left(\frac{\tau_{\text{motor}} \cdot N_{\text{gear}} \cdot \eta}{R}, \mu F_z \right) \quad (102)$$

Advantages:

- Simple, fast computation
- No additional state variables
- Stable at all speeds
- Captures traction limiting

Limitations:

- Cannot model wheel spin ($\omega \gg V_x/R$)
- Cannot model wheel lock ($\omega = 0$ while $V_x > 0$)
- Slip ratio is estimated, not computed
- No TCS/ABS testing capability

10.2 Hybrid Model

Full causality: $\tau_{\text{motor}} \rightarrow \omega \rightarrow \sigma_x \rightarrow F_x$

Advantages:

- Physical wheel spin and lock behavior
- Natural slip development
- Full Dugoff slip-force curve (including peak and sliding)
- TCS/ABS algorithm testing
- Realistic launch and braking dynamics

Limitations:

- Additional state variables (4 wheel speeds)
- Requires smaller timestep for stability
- More complex initialization

10.3 Model Selection Guide

Use Case	Torque-Only	Hybrid
Path following	✓	✓
Energy estimation	✓	✓
Traction limiting	✓	✓
Wheel spin detection		✓
ABS/TCS development		✓
Stability control		✓
Loss of control		✓

Table 2: Model selection based on use case

11 Implementation Algorithm

11.1 Torque-Only Algorithm

Step 1: Compute Normal Loads

1. Static distribution from geometry
2. Load transfer: $\Delta F_z = ma_x h_{cg}/L$
3. Per-wheel loads

Step 2: Compute Demanded Force

$$F_{x,\text{demanded}} = \frac{\tau_{\text{motor}} \cdot N_{\text{gear}} \cdot \eta}{R} \quad (103)$$

Step 3: Apply Friction Limit

$$F_x = \min(|F_{x,\text{demanded}}|, \mu F_z) \cdot \text{sgn}(F_{x,\text{demanded}}) \quad (104)$$

Step 4: Update Vehicle Dynamics

$$a = \frac{\sum F_x - F_{\text{drag}} - F_{\text{roll}}}{m} \quad (105)$$

11.2 Hybrid Algorithm

Step 1: Compute Normal Loads (same as torque-only)

Step 2: Compute Slip Ratios from Current Wheel Speeds

$$\sigma_x = \frac{\omega R - V_x}{\max(|V_x|, |\omega R|, V_{\min})} \quad (106)$$

Step 3: Compute Tire Forces (Dugoff)

$$\lambda = \frac{\mu F_z}{2\sqrt{(C_x \sigma_x)^2 + (C_y \sigma_y)^2}} \quad (107)$$

$$f(\lambda) = \begin{cases} (2 - \lambda)\lambda & \lambda < 1 \\ 1 & \lambda \geq 1 \end{cases} \quad (108)$$

$$F_x = C_x \sigma_x f(\lambda) \quad (109)$$

Step 4: Update Wheel Dynamics

$$\omega[k+1] = \omega[k] + \frac{\Delta t}{I_w}(\tau_{\text{drive}} - \tau_{\text{brake}} \cdot \text{sgn}(\omega) - F_x R) \quad (110)$$

Step 5: Update Vehicle Dynamics

$$V_x[k+1] = V_x[k] + \frac{\Delta t}{m}(\sum F_x - F_{\text{drag}} - F_{\text{roll}}) \quad (111)$$

11.3 Complete Discrete-Time State Update

$$\omega_i[k+1] = \omega_i[k] + \Delta t \cdot \dot{\omega}_i \quad (\text{for each wheel } i) \quad (112)$$

$$V_x[k+1] = V_x[k] + \Delta t \cdot a_x \quad (113)$$

$$\dot{\psi}[k] = \frac{V_x[k]}{L} \tan(\delta[k]) \quad (\text{or from lateral dynamics}) \quad (114)$$

$$\psi[k+1] = \psi[k] + \Delta t \cdot \dot{\psi}[k] \quad (115)$$

$$x[k+1] = x[k] + \Delta t \cdot V_x[k] \cos(\psi[k]) \quad (116)$$

$$y[k+1] = y[k] + \Delta t \cdot V_x[k] \sin(\psi[k]) \quad (117)$$

12 Surface Friction Coefficients

12.1 Mining Haul Road Conditions

Surface	μ_{peak}	μ_{slide}
Dry pavement	0.85	0.70
Compact gravel	0.72	0.60
Loose gravel	0.55	0.45
Iron ore dust (dry)	0.45	0.35
Iron ore dust (wet)	0.30	0.22
Mud	0.25	0.18
Sand (loose)	0.40	0.32

Table 3: Friction coefficients for Pilbara mining conditions

12.2 Effect of Surface Friction

The surface friction μ directly affects:

1. **Maximum traction force:** $F_{x,\text{max}} = \mu F_z$
2. **Maximum acceleration:** $a_{\text{max}} = \mu g$ (flat road)
3. **Stopping distance:** $d_{\text{stop}} = \frac{v_0^2}{2\mu g}$
4. **Friction utilization:** Lower μ means earlier saturation ($\lambda < 1$)

13 Implementation Parameters

13.1 XCMG XDE320 Mining Truck

Parameter	Value	Unit
Mass (empty)	218,000	kg
Mass (loaded)	538,000	kg
Wheelbase L	6.3	m
Track width W	4.0	m
CG height h_{cg}	2.8	m
Tire radius R	1.93	m
Gear ratio N_{gear}	9.0	-
Drivetrain efficiency η	0.92	-
Motor max torque	145,000	N·m
Motor max power	2,100	kW

Table 4: Vehicle parameters

Parameter	Value	Unit
$C_{x,base}$	280,000	N
$C_{y,base}$	220,000	N
$F_{z,ref}$	100,000	N
I_w (front)	800	kg·m ²
I_w (rear)	1,200	kg·m ²

Table 5: Tire and wheel parameters

Parameter	Value	Unit
Maximum steering angle δ_{max}	35	deg
Steering rate limit $\dot{\delta}_{max}$	45	deg/s
Ackermann percentage α_{ack}	100	%
Speed limit start v_{start}	8	m/s
Speed limit end v_{end}	25	m/s
High-speed ratio r_{high}	0.35	-
Brake bias (front) β_{front}	0.40	-

Table 6: Steering and brake parameters

14 Validation and Benchmarks

14.1 Expected Behavior

Scenario 1: Acceleration on gravel ($\mu = 0.72$)

XCMG XDE320 empty (218 tons):

$$F_{z,rear} \approx 0.60 \times 218,000 \times 9.81 = 1,283,000 \text{ N} \quad (118)$$

$$F_{x,max} = 0.72 \times 1,283,000 = 924,000 \text{ N} \quad (119)$$

$$a_{max} = \frac{924,000}{218,000} = 4.2 \text{ m/s}^2 \quad (120)$$

Scenario 2: Acceleration on wet dust ($\mu = 0.30$)

$$F_{x,\max} = 0.30 \times 1,283,000 = 385,000 \text{ N} \quad (121)$$

$$a_{\max} = \frac{385,000}{218,000} = 1.8 \text{ m/s}^2 \quad (122)$$

Result: Acceleration limited to 1.8 m/s² on wet dust vs 4.2 m/s² on gravel.

14.2 Stopping Distance Comparison

For $v_0 = 50 \text{ km/h} = 13.9 \text{ m/s}$:

$$d_{\text{gravel}} = \frac{13.9^2}{2 \times 0.72 \times 9.81} = 13.7 \text{ m} \quad (123)$$

$$d_{\text{wet dust}} = \frac{13.9^2}{2 \times 0.30 \times 9.81} = 32.8 \text{ m} \quad (124)$$

Stopping distance increases 2.4× on wet dust compared to gravel.

14.3 Minimum Turn Radius Verification

At $\delta_{\max} = 35$ with $L = 6.3 \text{ m}$:

$$R_{\min} = \frac{6.3}{\tan(35)} = 9.0 \text{ m} \quad (125)$$

Inner wheel turn radius: $R_{\text{inner}} = 9.0 - 2.0 = 7.0 \text{ m}$

This matches the expected tight-turn capability for pit operations.

15 Future Enhancements

15.1 Short-Term (V2.5)

- ✓ Wheel rotational dynamics (this document)
- ✓ Hybrid torque-slip model
- ✓ Ackermann steering geometry
- Combined longitudinal-lateral friction circle

15.2 Medium-Term (V3)

- 3-DOF vehicle dynamics (add $V_y, \dot{\psi}$ as states)
- Oversteer/understeer simulation
- Loss of control scenarios
- Steering aligning torque feedback

15.3 Long-Term (V4)

- 6-DOF dynamics (roll, pitch, heave)
- Suspension dynamics
- Rollover detection and prevention
- Four-wheel steering option

16 Conclusion

This document presents a comprehensive vehicle dynamics model with:

Steering System:

- Ackermann geometry for individual wheel angles
- Configurable Ackermann percentage
- Speed-dependent steering limit
- Rate limiting for hydraulic systems

Tire Modeling:

- Torque-only model for simple traction limiting
- Hybrid torque-slip model with wheel dynamics
- Full Dugoff slip-force characteristic
- Friction circle constraint

Vehicle Dynamics:

- Load transfer during acceleration/braking
- Per-wheel normal load computation
- Force-based longitudinal dynamics
- Surface friction effects

This forms a solid foundation for autonomous mining vehicle simulation and control algorithm development.

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A Sensor Models

A.1 Sensor Noise Models

A.1.1 Gaussian White Noise

Additive zero-mean Gaussian noise:

$$y = x_{\text{truth}} + \mathcal{N}(0, \sigma^2) \quad (126)$$

where $\mathcal{N}(0, \sigma^2)$ is normal distribution with standard deviation σ .

Implementation: Box-Muller transform:

$$U_1, U_2 \sim \mathcal{U}(0, 1) \quad (\text{uniform random}) \quad (127)$$

$$Z_1 = \sqrt{-2 \ln U_1} \cos(2\pi U_2) \quad (128)$$

$$Z_2 = \sqrt{-2 \ln U_1} \sin(2\pi U_2) \quad (129)$$

where $Z_1, Z_2 \sim \mathcal{N}(0, 1)$.

Scaled noise: $\sigma Z_1 \sim \mathcal{N}(0, \sigma^2)$.

A.1.2 Random Walk

Brownian motion (Wiener process):

$$dx = \sigma_{\text{rw}} dW \quad (130)$$

where $dW \sim \mathcal{N}(0, dt)$ is Wiener increment.

Discrete implementation:

$$x_{k+1} = x_k + \sigma_{\text{rw}} \sqrt{\Delta t} \cdot \mathcal{N}(0, 1) \quad (131)$$

Example: GNSS position drift with $\sigma_{\text{rw}} = 0.1 \text{ m}/\sqrt{\text{s}}$.

A.1.3 Gauss-Markov Process

First-order Markov process (exponentially correlated noise):

$$\frac{db}{dt} = -\frac{1}{\tau} b + w(t) \quad (132)$$

where:

- b – bias state
- τ – correlation time constant (s)
- $w(t) \sim \mathcal{N}(0, \sigma_w^2)$ – white noise driving term

Discrete solution:

$$b_{k+1} = e^{-\Delta t/\tau} b_k + \sigma_b \sqrt{1 - e^{-2\Delta t/\tau}} \cdot \mathcal{N}(0, 1) \quad (133)$$

where σ_b is steady-state bias standard deviation.

Steady-state variance:

$$\sigma_b^2 = \frac{\tau \sigma_w^2}{2} \quad (134)$$

A.2 IMU Sensor Model

A.2.1 Gyroscope Model

Measured angular velocity:

$$\omega_{\text{meas}} = \omega_{\text{truth}} + b_{\text{gyro}} + n_{\text{gyro}} \quad (135)$$

where:

- ω_{truth} – true angular velocity (rad/s)
- b_{gyro} – time-varying bias (Gauss-Markov, $\tau = 1800 \text{ s}$)
- $n_{\text{gyro}} \sim \mathcal{N}(0, \sigma_{\text{gyro}}^2)$ – white noise ($\sigma = 0.1/\text{s}$)

Bias evolution:

$$b_{\text{gyro}, k+1} = \beta b_{\text{gyro}, k} + \sigma_b \sqrt{1 - \beta^2} \cdot \mathcal{N}(0, 1) \quad (136)$$

where $\beta = \exp(-\Delta t/\tau)$.

A.2.2 Accelerometer Model

Measured acceleration (body frame):

$$\mathbf{a}_{\text{meas}} = \mathbf{a}_{\text{truth}} + \mathbf{b}_{\text{accel}} + \mathbf{n}_{\text{accel}} \quad (137)$$

Truth values in body frame:

$$a_{x,\text{truth}} = \frac{dv}{dt} \quad (\text{longitudinal acceleration}) \quad (138)$$

$$a_{y,\text{truth}} = v \cdot \dot{\psi} \quad (\text{centripetal acceleration}) \quad (139)$$

$$a_{z,\text{truth}} = -g = -9.81 \text{ m/s}^2 \quad (\text{gravity, body Z up}) \quad (140)$$

Bias: Gauss-Markov with $\tau = 3600 \text{ s}$ (1 hour), $\sigma_b = 0.02 \text{ m/s}^2$.

White noise: $\sigma_{\text{accel}} = 0.05 \text{ m/s}^2$.

A.2.3 Gravity Compensation

Accelerometers measure **specific force** (excluding gravity). For a stationary vehicle:

$$\mathbf{a}_{\text{meas}} = -\mathbf{g}_{\text{body}} + \text{bias} + \text{noise} \quad (141)$$

where $\mathbf{g}_{\text{body}} = [0, 0, -9.81]^\top \text{ m/s}^2$ in body frame (Z-up).

During motion:

$$\mathbf{a}_{\text{meas}} = \mathbf{a}_{\text{vehicle}} - \mathbf{g}_{\text{body}} + \text{bias} + \text{noise} \quad (142)$$

A.3 GNSS Sensor Model

A.3.1 Position Measurement

Latitude and longitude with random walk drift:

$$\phi_{\text{meas}} = \phi_{\text{truth}} + d_\phi + \mathcal{N}(0, \sigma_\phi^2) \quad (143)$$

$$\lambda_{\text{meas}} = \lambda_{\text{truth}} + d_\lambda + \mathcal{N}(0, \sigma_\lambda^2) \quad (144)$$

where:

- d_ϕ, d_λ – random walk drift ($\sigma_{\text{rw}} = 0.1 \text{ m}/\sqrt{\text{s}}$)
- $\sigma_\phi = 2.0 \text{ m} / m_\phi$ – white noise (latitude)
- $\sigma_\lambda = 2.0 \text{ m} / m_\lambda$ – white noise (longitude)

A.3.2 Velocity Measurement (NED Frame)

North-East velocity:

$$v_{N,\text{meas}} = v_N + \mathcal{N}(0, 0.1^2) \quad (145)$$

$$v_{E,\text{meas}} = v_E + \mathcal{N}(0, 0.1^2) \quad (146)$$

where $v_N = v \sin(\psi)$ and $v_E = v \cos(\psi)$ from body velocity transformation.

A.3.3 Circular Error Probable (CEP)

The 50% probability error circle radius:

$$\text{CEP} = 0.59(\sigma_N + \sigma_E) \approx 1.2 \text{ m} \quad (147)$$

for $\sigma_N = \sigma_E = 2.0 \text{ m}$.

B Wheel Speed Encoder Model

B.0.1 Quantization Model

Encoder with N_{ticks} pulses per revolution:

$$\text{Resolution} = \frac{2\pi}{N_{\text{ticks}}} \text{ rad} \quad (148)$$

Quantized measurement:

$$\omega_{\text{quant}} = \left\lfloor \frac{\omega_{\text{truth}}}{\text{Resolution}} \right\rfloor \cdot \text{Resolution} \quad (149)$$

where $\lfloor \cdot \rfloor$ denotes rounding to nearest integer.

B.0.2 Total Encoder Noise

Combined quantization and sensor noise:

$$\omega_{\text{meas}} = \omega_{\text{quant}} + \mathcal{N}(0, \sigma_{\omega}^2) \quad (150)$$

where $\sigma_{\omega} = 0.5 \text{ rad/s}$ (white noise).

B.0.3 Per-Wheel Variation

Simulate tire pressure and slip variations:

$$\omega_i = \omega_{\text{meas}} \cdot (1 + u_i) \quad (151)$$

where $u_i \sim \mathcal{U}(-0.02, +0.02)$ is uniform random variable (2% variation).

B.1 Radar Sensor Model

B.1.1 Range Measurement

Simulated stationary target at fixed distance:

$$r_{\text{meas}} = r_{\text{truth}} + \mathcal{N}(0, \sigma_r^2) \quad (152)$$

where $r_{\text{truth}} = 50.0 \text{ m}$ and $\sigma_r = 0.2 \text{ m}$.

B.1.2 Doppler Velocity

Relative radial velocity (closing speed):

$$v_{\text{rel}} = v_{\text{ego}} \cos(\theta) \quad (153)$$

where θ is angle between vehicle heading and line-of-sight to target.

For target dead ahead ($\theta = 0$):

$$v_{\text{meas}} = v_{\text{truth}} + \mathcal{N}(0, \sigma_v^2) \quad (154)$$

where $\sigma_v = 0.1 \text{ m/s}$.

B.1.3 Angle Measurement

Azimuth angle to target:

$$\alpha_{\text{meas}} = \alpha_{\text{truth}} + \mathcal{N}(0, \sigma_{\alpha}^2) \quad (155)$$

where $\alpha_{\text{truth}} = 0$ (dead ahead) and $\sigma_{\alpha} = 0.5$.

C Coordinate Systems and Transformations

C.1 Reference Frames

C.1.1 Global Inertial Frame (World)

Fixed reference frame with origin at simulation start:

- X – East (or initial vehicle forward direction)
- Y – North (or initial vehicle left direction)
- Z – Up (vertical, right-hand rule)

Vehicle state in global frame: (x, y, ψ) where ψ is heading angle (yaw) measured counter-clockwise from X -axis.

C.1.2 Vehicle Body Frame

Moving frame attached to vehicle with origin at rear axle center:

- x_b – Forward (longitudinal)
- y_b – Left (lateral)
- z_b – Up (vertical)

Vehicle velocity in body frame: v (longitudinal speed, m/s).

C.1.3 North-East-Down (NED) Frame

Navigation frame used by GNSS:

- N – North
- E – East
- D – Down (toward Earth center)

C.2 Coordinate Transformations

C.2.1 Body to Global

Position rate of change:

$$\dot{x} = v \cos(\psi) \quad (156)$$

$$\dot{y} = v \sin(\psi) \quad (157)$$

Rotation matrix from body to global:

$$\mathbf{R}_b^g = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (158)$$

C.2.2 Body to NED

Velocity transformation:

$$v_N = v \sin(\psi) \quad (159)$$

$$v_E = v \cos(\psi) \quad (160)$$

$$v_D = 0 \quad (\text{ground vehicle}) \quad (161)$$

C.2.3 Local XY to WGS84

Flat Earth approximation for short distances (< 10 km):

$$\text{meters per degree latitude: } m_\phi = 111,320 \text{ m} \quad (162)$$

$$\text{meters per degree longitude: } m_\lambda = 111,320 \cdot \cos(\phi_0) \text{ m} \quad (163)$$

Position conversion:

$$\phi = \phi_0 + \frac{y}{m_\phi} \quad (164)$$

$$\lambda = \lambda_0 + \frac{x}{m_\lambda} \quad (165)$$

where (ϕ_0, λ_0) is the origin in WGS84 coordinates (latitude, longitude in degrees).

D Bicycle Model Kinematics

D.1 Model Assumptions

1. **No slip:** Pure rolling condition (wheels do not slide)
2. **Planar motion:** Vehicle remains on flat ground (no pitch/roll)
3. **Point mass:** Neglect rotational inertia about vertical axis
4. **Rear-axle reference:** Kinematic center at rear axle midpoint

D.2 Kinematic Equations

The bicycle model reduces a four-wheeled vehicle to two wheels separated by wheelbase L :

$$\begin{aligned} \dot{x} &= v \cos(\psi) \\ \dot{y} &= v \sin(\psi) \\ \dot{\psi} &= \frac{v}{L} \tan(\delta) \end{aligned} \quad (166)$$

where:

- (x, y) – position of rear axle center in global frame (m)
- ψ – heading angle (rad)
- v – longitudinal velocity (m/s)
- δ – virtual bicycle steering angle (rad)
- L – wheelbase (m)

D.3 Derivation of Yaw Rate

Step 1: Front wheel velocity perpendicular to wheel heading (no-slip):

$$v_f = \frac{v}{\cos(\delta)} \quad (167)$$

Step 2: Lateral component of front wheel velocity:

$$v_{f,\text{lateral}} = v_f \sin(\delta) = v \tan(\delta) \quad (168)$$

Step 3: Angular velocity about rear axle (instantaneous center of rotation):

$$\dot{\psi} = \frac{v_{f,\text{lateral}}}{L} = \frac{v \tan(\delta)}{L} \quad (169)$$

D.4 Turning Radius

From the yaw rate equation:

$$R = \frac{L}{\tan(\delta)} \quad (170)$$

For small angles ($|\delta| < 0.2 \text{ rad} \approx 11^\circ$):

$$\tan(\delta) \approx \delta \implies R \approx \frac{L}{\delta} \quad (171)$$

D.5 Discrete-Time Integration

Using Euler forward integration with timestep Δt :

$$\psi_{k+1} = \psi_k + \Delta t \cdot \frac{v_k}{L} \tan(\delta_k) \quad (172)$$

$$x_{k+1} = x_k + \Delta t \cdot v_k \cos(\psi_k) \quad (173)$$

$$y_{k+1} = y_k + \Delta t \cdot v_k \sin(\psi_k) \quad (174)$$

Stability condition: To prevent integration instability:

$$\Delta t < \frac{L}{v_{\max} \tan(\delta_{\max})} \quad (175)$$

For $L = 2.8 \text{ m}$, $v_{\max} = 60 \text{ m/s}$, $\delta_{\max} = 35^\circ \approx 0.61 \text{ rad}$:

$$\Delta t < 0.067 \text{ s} = 67 \text{ ms} \quad (176)$$

Chosen timestep: $\Delta t = 10 \text{ ms}$ provides $6.7\times$ safety margin.

E Ackermann Steering Geometry

E.1 Problem Statement

Real vehicles have two front wheels that must turn at different angles to maintain pure rolling (no tire scrubbing). All wheels must rotate about a common **instantaneous center of rotation (ICR)**.

E.2 Geometric Configuration

Parameter	Symbol
Wheelbase	L
Track width (front)	W
Virtual steering angle	δ
Turning radius (rear axle to ICR)	R

E.3 Derivation

Step 1: Turning radius from bicycle model:

$$R = \frac{L}{\tan(\delta)} \quad (177)$$

Step 2: Inner and outer wheel radii:

$$R_{\text{inner}} = R - \frac{W}{2} \quad (178)$$

$$R_{\text{outer}} = R + \frac{W}{2} \quad (179)$$

Step 3: Front wheel angles from geometry:

$$\delta_{\text{inner}} = \arctan\left(\frac{L}{R_{\text{inner}}}\right) = \arctan\left(\frac{L}{R - W/2}\right) \quad (180)$$

$$\delta_{\text{outer}} = \arctan\left(\frac{L}{R_{\text{outer}}}\right) = \arctan\left(\frac{L}{R + W/2}\right) \quad (181)$$

E.4 Turn Direction Convention

For **left turn** ($\delta > 0$):

$$\delta_{FL} = \arctan\left(\frac{L}{R - W/2}\right) \quad (\text{inner, larger angle}) \quad (182)$$

$$\delta_{FR} = \arctan\left(\frac{L}{R + W/2}\right) \quad (\text{outer, smaller angle}) \quad (183)$$

For **right turn** ($\delta < 0$):

$$\delta_{FR} = \arctan\left(\frac{L}{R - W/2}\right) \quad (\text{inner, larger magnitude}) \quad (184)$$

$$\delta_{FL} = \arctan\left(\frac{L}{R + W/2}\right) \quad (\text{outer, smaller magnitude}) \quad (185)$$

E.5 Small Angle Approximation

For $|\delta| \ll 1$:

$$\delta_{\text{inner}} \approx \frac{L}{R - W/2} = \delta \left(1 + \frac{W}{2R}\right) \quad (186)$$

$$\delta_{\text{outer}} \approx \frac{L}{R + W/2} = \delta \left(1 - \frac{W}{2R}\right) \quad (187)$$

F Longitudinal Dynamics

F.1 Force Balance

Newton's second law in longitudinal direction:

$$m \frac{dv}{dt} = F_{\text{motor}} - F_{\text{brake}} - F_{\text{drag}} - F_{\text{roll}} \quad (188)$$

where m is vehicle mass (kg) and v is longitudinal velocity (m/s).

F.2 Motor Force

Electric motor torque converted to wheel force:

$$F_{\text{motor}} = \frac{T_{\text{motor}} \cdot N_{\text{gear}} \cdot \eta_{\text{drivetrain}}}{r_{\text{wheel}}} \quad (189)$$

where:

- T_{motor} – motor shaft torque (Nm)
- N_{gear} – gear ratio (dimensionless)
- $\eta_{\text{drivetrain}}$ – drivetrain efficiency (typically 0.92)
- r_{wheel} – effective wheel radius (m)

F.3 Power Constraint

Motors are torque-limited at low speed and power-limited at high speed:

$$P_{\text{motor}} = T_{\text{motor}} \cdot \omega_{\text{motor}} \quad (190)$$

Motor angular velocity:

$$\omega_{\text{motor}} = \frac{v}{r_{\text{wheel}}} \cdot N_{\text{gear}} \quad (191)$$

Maximum torque at given speed:

$$T_{\text{motor}}^{\text{max}}(v) = \frac{P_{\text{motor}}^{\text{max}}}{\omega_{\text{motor}}(v)} = \frac{P_{\text{motor}}^{\text{max}} \cdot r_{\text{wheel}}}{v \cdot N_{\text{gear}}} \quad (192)$$

F.4 Aerodynamic Drag

Quadratic drag force:

$$F_{\text{drag}} = \frac{1}{2} \rho C_d A v^2 = C_{\text{drag}} v |v| \quad (193)$$

where:

- ρ – air density (1.225 kg/m³ at sea level)
- C_d – drag coefficient (dimensionless, 0.25–1.2 depending on vehicle)
- A – frontal area (m²)
- $C_{\text{drag}} = \frac{1}{2} \rho C_d A$ – combined drag constant

F.5 Rolling Resistance

Simplified constant friction model:

$$F_{\text{roll}} = C_{\text{roll}} \cdot \text{sign}(v) \quad (194)$$

where C_{roll} is rolling resistance force magnitude (N).

More detailed model:

$$F_{\text{roll}} = \mu_{\text{roll}} \cdot m \cdot g \quad (195)$$

where μ_{roll} is rolling resistance coefficient (0.01–0.03 for asphalt).

F.6 Friction Braking

Brake force proportional to pedal input:

$$F_{\text{brake}} = \frac{p_{\text{brake}}}{100} \cdot \frac{T_{\text{brake}}^{\text{max}}}{r_{\text{wheel}}} \quad (196)$$

where $p_{\text{brake}} \in [0, 100]$ is brake pedal percentage.

F.7 Velocity Update

Acceleration:

$$a = \frac{F_{\text{motor}} - F_{\text{brake}} - F_{\text{drag}} - F_{\text{roll}}}{m} \quad (197)$$

Discrete velocity update:

$$v_{k+1} = v_k + \Delta t \cdot a_k \quad (198)$$

Zero-crossing logic: Prevent oscillation near standstill:

$$\text{if } \text{sign}(v_{k+1}) \neq \text{sign}(v_k) \text{ and } |v_{k+1}| < \epsilon_v, \text{ then } v_{k+1} = 0 \quad (199)$$

where $\epsilon_v = 0.3$ m/s is velocity threshold.

G Lateral Dynamics and Friction Limits

G.1 Centripetal Acceleration

During cornering, lateral acceleration is:

$$a_{\text{lat}} = v \cdot \dot{\psi} = \frac{v^2}{L} \tan(\delta) \quad (200)$$

G.2 Friction Circle Constraint

Maximum lateral acceleration limited by tire-road friction:

$$|a_{\text{lat}}| \leq \mu g \quad (201)$$

where:

- μ – lateral friction coefficient (0.1–0.9 depending on surface)
- $g = 9.81$ m/s² – gravitational acceleration

G.3 Understeer Implementation

When friction limit exceeded, scale yaw rate:

$$\dot{\psi}_{\text{actual}} = \dot{\psi}_{\text{commanded}} \cdot \min\left(1, \frac{\mu g}{|a_{\text{lat}}|}\right) \quad (202)$$

H Validation and Verification (Kinematics/Sensor Doc)

H.1 Statistical Tests

H.1.1 Chi-Squared Goodness of Fit

Test if noise distribution matches Gaussian:

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \quad (203)$$

where O_i are observed counts and E_i are expected counts in histogram bins.

H.1.2 Allan Variance

Characterize gyroscope noise:

$$\sigma_A^2(\tau) = \frac{1}{2(N-1)} \sum_{i=1}^{N-1} (\bar{\omega}_{i+1}(\tau) - \bar{\omega}_i(\tau))^2 \quad (204)$$

where $\bar{\omega}_i(\tau)$ is average over interval τ .

H.2 Coordinate Transform Verification

Round-trip test: Body $\rightarrow$ NED $\rightarrow$ Body should preserve magnitude:

$$v = \sqrt{v_N^2 + v_E^2} \quad (\text{Pythagorean theorem}) \quad (205)$$

Gravity check: Stationary vehicle should measure:

$$\mathbf{a}_{\text{meas}} = [0, 0, -9.81]^\top + \text{small noise/bias} \quad (206)$$

I Conclusion (Kinematics/Sensor Doc)

This mathematical framework provides the foundation for:

- High-fidelity vehicle dynamics simulation
- Realistic sensor noise modeling for ML/AI training
- Coordinate transformations between body, NED, and WGS84 frames
- Extended Kalman Filter (EKF) development and tuning
- Sensor fusion algorithm validation

All models are validated against industry standards and real sensor specifications.

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